How Sundial makes a forecast

Tsinghua University (THUML) · 128M parameters · decoder-only transformer with a flow-matching head · zero-shot

1. One series at a time

Sundial reads only the target's own past. A ? is a value not known yet.

past *future*

Target temperature

Covariates are not used.

Five metal prices become five separate forecasts.

2. Scale, then patches of 16

The context is scaled by its mean μ and standard deviation σ. Each patch and its seen flags go through a residual MLP.

$$\tilde{x} = \frac{x - \mu}{\sigma}$$

value

seen?

one patch = 16 steps

residual MLP

one token: 768 numbers

3. Twelve causal blocks

Each patch reads only itself and earlier patches. LayerNorm in front, shortcut around.

earlier → later

earlier patches only

Self-attention: causal, 12 heads, rotary positions (RoPE)

Feed forward: SwiGLU, 768 → 3,072 → 768

4. Paths drawn from noise

The last token steers a small network that turns random noise into a **720-step path** in 50 small steps (flow matching).

noise → path, step by step

median

sample paths

A fresh noise draw gives another path; the median of the paths is the point forecast.

A flow-matching head draws whole forecast paths instead of a fixed set of quantiles.

12 blocks · width 768 · 12 heads · patches of 16 · up to 720 steps per pass · 50 flow steps

In the notebooks: last 2,048 days or full monthly history · each metal on its own · no covariates · 20 sample paths